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Carbon nanotubes modified graphene hybrid Aerogel-based composite phase change materials for efficient thermal storage

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ABSTRACT

Phase change materials (PCMs) have attracted more and more attention in the field of latent heat storage. In order to solve the leakage behavior and improve heat transfer performance of polyethylene glycol (PEG), a method of self-assembled hybrid aerogel filled with carbon nanotubes (CNTs) combined with reduced graphene oxide (rGO) nanosheets is proposed herein. The as-prepared CNTs-graphene hybrid aerogel (CGA) with three-dimensional porous structure can be used as a highly adsorptive and stable carrier for PCM. After the vacuum impregnation process, a high-performance PEG/CGA composite PCMs (c-PCMs) is obtained, which exhibits an outstanding PEG loading ratio of 97.6 %. Moreover, after 1000 repeated heating/cooling cycles, the H_M and H_S value of PEG/CGA c-PCMs remained 93.44 % and 92.84 %. Meanwhile the PEG/CGA c-PCMs remains intact and the loading ratio could still maintain as high as 96.4 %, which proved that PEG/CGA c-PCMs presented reliable thermal cycling stability. The PEG/CGA c-PCMs only takes 603 s to raise from 30 ℃ to 70 ℃, which is much faster than that of pure PEG (813 s) and PEG/GA composite (727 s). And The thermal conductivity of PEG/CGA(0.586 W/(m·K)) was 186 % times than that of pure PEG(0.314 W/(m·K)).The introduction of CNTs can not only enhance the adsorption capacity of c-PCMs and alleviate the leakage of PEG, but also improve the heat transfer performance of c-PCMs. These results specify that PEG/CGA c-PCMs can be used as remarkable and stable energy storage materials for environmental thermal management systems.

1. Introduction

As various countries and regions around the world have proposed "carbon neutrality" or "zero carbon" goals, "green and low carbon" and "sustainable development" have become an international consensus [1]. This has fostered the development of sustainable and renewable energies, among which solar energy is widely regarded as the most important renewable energy source [2]. But the potential of solar energy is limited, mainly due to its intermittency, regional variability and unpredictability [3,4]. The unique properties of phase change materials (PCMs) were precisely solved this limitation, which could store a large amount of thermal energy through the phase change process and be used as a suitable medium for solar thermal energy conversion and storage [5,6]. Currently, PCMs including inorganic and organic PCMs, which have been used in many fields such as building energy conservation, thermal insulation, smart textiles, waste heat recovery and solar heating systems [7–9]. Compared with inorganic PCMs, affordable organic PCMs offer the benefits of moderate phase change enthalpy, wide melting temperature, and well-behaved chemical and thermal stability, which have been extensively investigated [7,10,11]. Among the various organic PCMs, polyethylene glycol (PEG) presents the advantages of high latent heat of phase change, low cost, non-toxicity, well-behaved chemical and thermal stability, etc., which makes it a promising solid–liquid organic PCM. However, the main imperfection such as the leak problem and the poor heat transfer during thermal storage and release process still limit its further application [12–16].

To tackle these two problems, some researchers have concentrated on improving the shape stability and heat transfer performance of PCMs. For solving the leak problem, the common method is to encapsulate PCMs into the matrix material to construct a shape-stabilized PCMs (ss-PCMs). The matrix materials with large specific surface area and excellent stability can control the fluidity of organic PCMs through physical interaction, thereby preventing the leak problem of PCMs [17–20]. For example, graphene aerogel (GA) [3,21–24], metal foam [25–30], porous carbon [12,13,31,32] and porous mineral materials [14,15,33,34] have

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been used as the matrix materials of PCMs. Among these, graphene aerogels (GA) formed by self-assembly of graphene oxide (GO) have various characteristics of conventional aerogels, such as porous three-dimensional structure, low density, high porosity, large specific surface area, high adsorption capability excellent thermal stability and so on [35,36], which can be used as a carrier for PEG leakage prevention. Additionally, the porous three-dimensional carbon framework of GA can provide heat transfer channel and further improve thermal conductivity. These characteristics are of great significance and have a tremendous application prospect as a carrier for PEG-based PCMs [3,21–24].

It is necessary to fully exploit the unique advantages of GA to prepare high-performance composite PCMs (c-PCMs) and explore their heat storage properties. However, the research on GA on PCMs is insufficient. Zheng et al. [37] encapsulated paraffin wax in a carrier composed of graphene aerogel (GA) and copper foam (CF) to produce a novel and stable c-PCMs. Its light-to-heat conversion rate is as high as 97 %, and its thermal conductivity is 9 times that of pure paraffin. Kong et al. [38] prepared the L-gGA2800 skeleton by the sol–gel method and high temperature heat treatment at 2800 °C and composited with paraffin to prepare the L-gGA2800/PA c-PCMs. The L-gGA2800/PA exhibited a high thermal conductivity of 0.92 W/m·K following encapsulation of paraffin with 3D aerogel as the skeleton. To sum up and analyse these research findings, there remain three aspects for further improvement. First, the loading ratio for PCMs they have prepared is approximately 80 % to 90 %, which is not high enough. For the same volume of c-PCMs, there is still scope to improve the capacity and efficiency of thermal storage. Secondly, in the cyclic stability test of the samples, the number of cycles was inadequate. If prepared c-PCMs are to be used in practice for an extended period, more cycles need to be defined to check their stability. In addition, even though the materials they have prepared have a high thermal conductivity compared to pure PCM, there is still a great deal of room for improvement over carbon materials with high intrinsic thermal conductivity. Therefore, it is necessary to develop c-PCMs which can meet the requirements of high loading ratio and excellent cycle stability by optimizing the structure of the GA.

Herein, we introduced the CNTs to manufacture CGA with the three-dimensional carbon porous structure, and then combined it with PEG to form PEG/CGA c-PCMs. The c-PCMs have presented multiple advantages. Firstly, due to the capillary and superficial tension of CNTs, the loading ratio and the thermal storage capacity are considerably improved. CNTs are combined in a crosslinked state and engaged between rGO nanosheets, which has increased the wall thickness of GA and enhanced the shape and cycle stability of the c-PCMs. Furthermore, the introduction of CNTs has enriched and shortened the thermal conduction path of the carrier, thereby improving the thermal conduction capacity of the c-PCMs. The CGA with three-dimensional porous structure has excellent adsorption performance for PEG, and the synthesized PEG/CGA c-PCMs has an excellent PEG loading ratio of 97.6 %. The PEG/CGA c-PCMs also exhibited outstanding cycling stability, the H_M and H_S value of PEG/CGA c-PCMs remained 93.44 % and 92.84 % after 1000 repeated heating/cooling cycles. Meanwhile the PEG/CGA c-PCMs remains intact and the loading ratio could still maintain as high as 96.4 % which is a decrease of only 1.2 % compared to the virgin ones. These results proved that PEG/CGA c-PCMs presented reliable thermal cycling stability. In addition, the time required for the PEG/CGA c-PCMs to increase from 30 °C to 70 °C was 603 s, a decline of 25.8 % and 17.1 % relative to pure PEG (813 s) and pure PEG/GA (727 s). And the thermal conductivity of PEG/CGA was 186 % times than that of pure PEG. These results revealed that the synthesized PEG/CGA c-PCMs have excellent loading ratio and outstanding cycling stability as well as great heat transfer performance. Consequently, it adds new possibilities for important and stable energy storage materials in thermal management systems of the environment.

2. Experiment

2.1. Materials

Polyethylene glycol (Xi Long Scientific Co Ltd.) with an average molecular weight of 4000 (PEG-4000, AR), CNTs (Aladdin Industrial Corporation) with 0.5–2 μm length, and GO (Suzhou Carbon Feng Graphene Technology Co Ltd) were used in this study.

2.2. CGA and GA preparation process

The schematic process diagram for preparing the CGA is presented in Fig. 1(a). Firstly, 60 mg of GO and various CNTs weights (CNTs account for 10 wt%, 30 wt%, 50 wt%, 70 wt% of the total weights) were added to 30 mL of deionized water to form a water suspension by ultrasonic stirring. Subsequently, the uniform dispersion was fed into a 50 mL high-pressure reactor and placed in a 200 °C oven for 6 h. During the hydrothermal reaction, hybrid hydrogel was formed through the assembly of high conductivity reduced graphene oxide (rGO) nanosheets. Then the hydrogels were transferred to lyophilization for 48 h to fabricate CGA-10, CGA-30, CGA-50 and CGA-70 which are carriers with a three-dimensional porous framework structure. The preparatory process for the GA was identical to that for the CGA, but without the CNTs.

2.3. Preparation of the PEG/CGA c-PCMs and PEG/GA c-PCMs

As illustrated in Fig. 1(b), the PEG/CGA c-PCMs were synthesized via a vacuum impregnation process. First, the excessed solid PEG was mixed with CGA and were heated at 80 °C for 3 h in a vacuum oven ensure the solid PEG was entirely melted which made PEG much easier to combine with the carrier. The remaining PEG on the CGA surface was removed through heating in an oven at 80 °C. Heating steps were carried out until a PEG was not observed in the filter paper. Finally, the PEG/CGA c-PCMs were obtained. Moreover, as the reference sample, PEG/GA c-PCMs were prepared via the same steps.

2.4. Analysis method

The micro-morphology of GA, CGA, PEG/GA and PEG/CGA c-PCMs were observed by scanning electronic microscopy (SEM, Model Zeiss Supra-55). The chemical compatibility of c-PCMs were tested via Fourier transform infrared spectroscopy (FT-IR, Model Per-kin Elmer Spectrum 100) and X-ray diffraction (XRD, Model AXS D8 Advance). The measurement of FT-IR ranged from 400 to 4500 cm^{-1} and the XRD varied from 10° to 80° at a scanning rate of 8°/min. Moreover, the phase change temperature and enthalpy were explored using Simultaneous Thermal Analysis (STA-TGA/DSC, Model NETZSCH STA449F3), which was implemented by heating and cooling from 0 °C to 100 °C at a rate of 10 °C/min under a nitrogen atmosphere. Thermo-gravimetric analysis (TGA) was conducted to investigate the thermal properties and stability of c-PCMs. Derivative thermo-gravimetric (DTG) measurements were conducted to investigate the thermal stability and decomposition properties of the c-PCMs. The heat transfer performance of PEG/CGA c-PCMs were observed by multi-recording data loggers and infrared thermal imagers. The thermal conductivity value was measured using a Xenon flash tester (Model LINSEIS XFA 500).

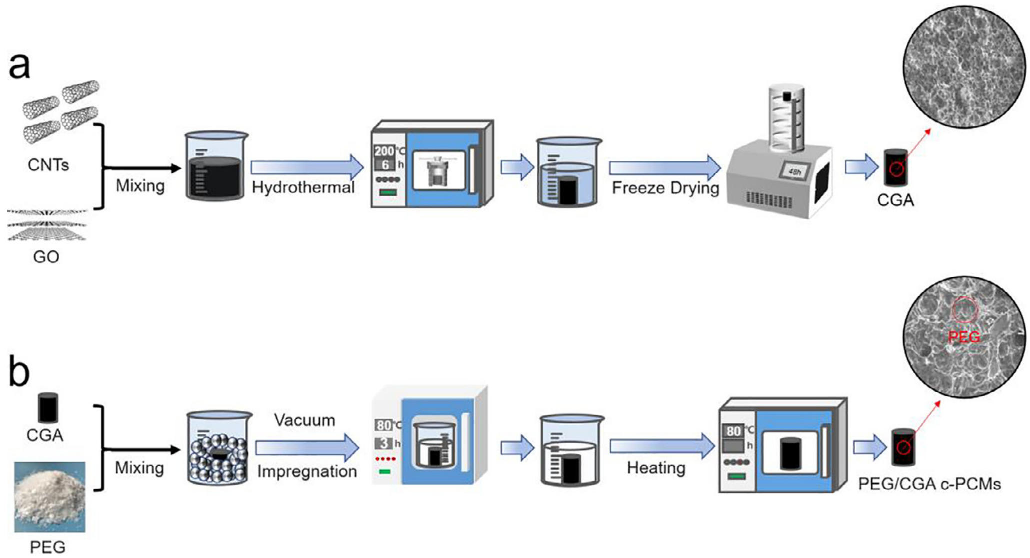

Fig. 1. (a-b) Schematic process for the preparation of CGA and PEG/CGA c-PCMs.

3. Results and discussion

3.1. Morphology and microstructure of CGA and PEG/CGA c-PCMs

The morphology of CGA-10, CGA-30, CGA-50 and CGA-70 heated at 80 °C are shown in Fig. 2(a-c). The results indicate that the shape of the c-PCMs has become unstable with the progressive increase in CNTs content. Under the same hydrothermal temperature and time, the original shape of CGA-30, CGA-50 and CGA-70 c-PCMs could not be maintained. The surface of CGA-30, CGA-50 and CGA-70 c-PCMs have collapsed to different degrees. Of these, the surface of CGA-70 c-PCMs has collapsed most seriously, and collapsed debris can be clearly seen from the morphology image. Only CGA-10 has retained its shape unchanged. The reason for this phenomenon may be that in the CGA three-dimensional porous structure, with the increase of CNTs content, the relative content of graphene nanosheets decreased. As a result, the π-π bond between graphene nanosheets decreased, meanwhile the excessed CNTs couldn't playact the role as a framework alone, thereby reduced the stability of the CGA structure and caused the shape collapsed.

According to the literature [39], the loading ratio of PEG in c-PCMs could be calculated by equation (1):

$$\eta = (W_A - W_B)/W_A \times 100\% \tag{1}$$

where W_B is the weight of the CGA before vacuum impregnation, W_A is the weight of PEG/CGA c-PCMs formed by vacuum impregnation of CGA and PEG, η presents the loading ratio of the PEG in c-PCMs. All results were summarized in Table 1. It can be seen that with the increase of CNTs content in CGA, the loading ratio of PEG gradually decreases. This is related to the structural stability of the CGA mentioned above. With the increase of CNTs content, the structural stability of CGA gradually decreases, which reduced the adsorption performance of the carrier for PEG. The result is that the PEG loading ratio decreased with the increase of CNTs content. Accordingly, it was decided to continue the search on CGA-10.

SEM was used to observe the porous structure of the CGA-10. Fig. 3(a) displays that the CGA-10 exhibits an interconnected,

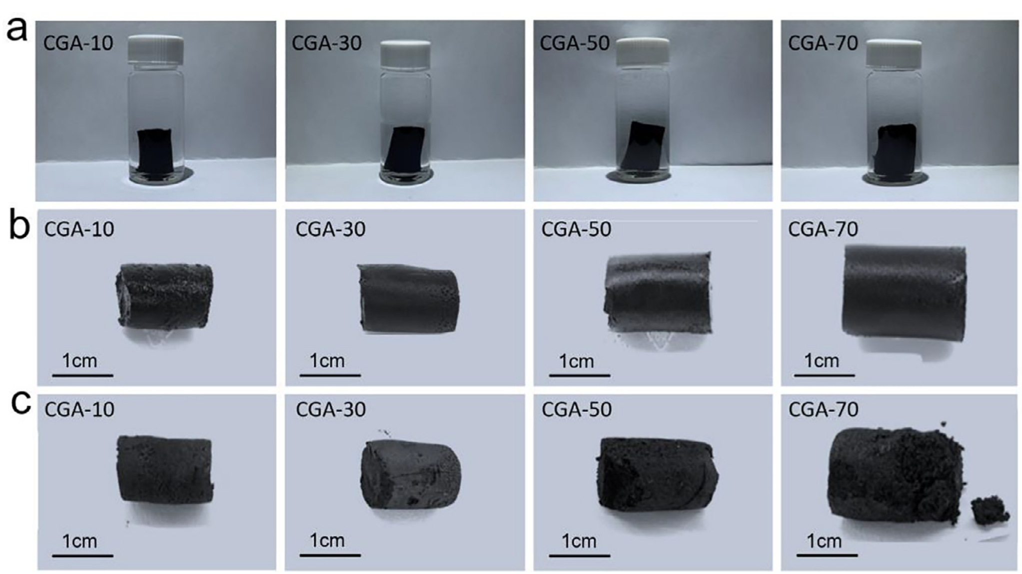

Fig. 2. The morphology of CGA-10, CGA-30, CGA-50 and CGA-70: (a) After hydrothermal, (b) Before impregnation, (c) After heating.

Table 1
PEG loading ratio of CGA with different CNTs contents.

Samples	W_B (g)	W_A (g)	Loading Ratio (wt%)
PEG/CGA-10	0.0325 g	1.3908 g	97.6
PEG/CGA-30	0.0451 g	0.7274 g	93.8
PEG/CGA-50	0.0597 g	0.6707 g	91.1
PEG/CGA-70	0.0694 g	0.5932 g	88.3

three-dimensional porous framework strcture of randomly oriented sheets with continuous open macropores. The rGO nanosheets separated the CGA-10 into compartmental volumes in three-dimensional, and the CNTs have supported and connected the rGO nanosheets. Fig. 3(b) presents a large number of cross-linked CNTs are combined between the rGO nanosheets. These CNTs served as supports to the walls of compartment, tightly connected the rGO nanosheets and extended outward to connect other rGO nanosheets. Thus, a three-dimensional and interconnected porous structure was formed, which not only enhanced the stability among the rGO nanosheets but also the heat transfer performance.

The internal structures of GA and CGA-10 have been observed by TEM. The TEM image (Fig. 3(c)) presents that the rGO nanosheets without CNTs were in the form of lamellae. TEM image (Fig. 3(d)) displays the quantity of CNTs with lengths of several micrometers were well distributed and adhered to the rGO nanosheets which were ultra-large, flat and complete in CGA-10. It can be seen that there were CNTs stacked on top of rGO nanosheets at the edges of the folds and curls and merged into the gaps between the different rGO nanosheets, which indicates that CNTs could effectively increase the stability of three-dimensional porous framework structure.

Therefore, the intact shape of CGA-10 should be attributed to the support of CNTs that were decumbent and interlocking among the rGO nanosheets to improve the stability and thickening the rGO nanosheets to prevent the collapse of interspaces. For CGA-30, CGA-50 and CGA-70 c-PCMs, with the incremental increase in CNTs addition content, the relative content of rGO nanosheets in the system decreased and rGO nanosheets tend to stretched and exposed the nanosheet as a single layer in a three-dimensional structure [40]. Therefore reduced the interaction between rGO nanosheets and structural stability of CGA. As a result, CGA-30, CGA-50 and CGA-70 c-PCMs suffered varying degrees of shape collapse.

After all, based on the above results, the optimal CNTs doping ratio has been determined to be 10 wt% by weight (the mass fraction of CNTs in CGA is 10 wt%). In the following part of this manuscript, the CGA refers to hybrid aerogel with CNTs doping ratio of 10 wt%.

3.2. Micromorphology and chemical compatibility of PEG/CGA c-PCMs

The morphology of PEG/CGA c-PCMs was investigated by SEM. As Compared with the microstructure of CGA (Fig. 3(a-b)), the interconnected and three-dimensional porous structure of PEG/CGA c-PCMs was not apparent and the pores of PEG/CGA c-PCMs was filled with solid PEG (Fig. 4(a)). From the SEM image (Fig. 4(b)) with a higher magnification, it is evident that the surface of the PEG/CGA c-PCMs was uniformly covered by the smooth layer of PEG and numerous cross-linked CNTs were filled between the rGO nanosheets in the cross section. The CNTs filled between the rGO nanosheets and enhanced the mechanical strength of the PEG/CGA c-PCMs during the phase transition process and has maintained the integrity of the PEG/CGA c-PCMs. Furthermore, the high capillary force and surface tension of these CNTs improved the CGA's adsorption capacity to PEG and reduced the leakage of PEG. Meanwhile the high thermal conductivity of CNTs enhanced the heat transfer performance of PEG/CGA c-PCMs. To explore the compatibility of CNTs, GA and PEG, FT-IR and XRD were used

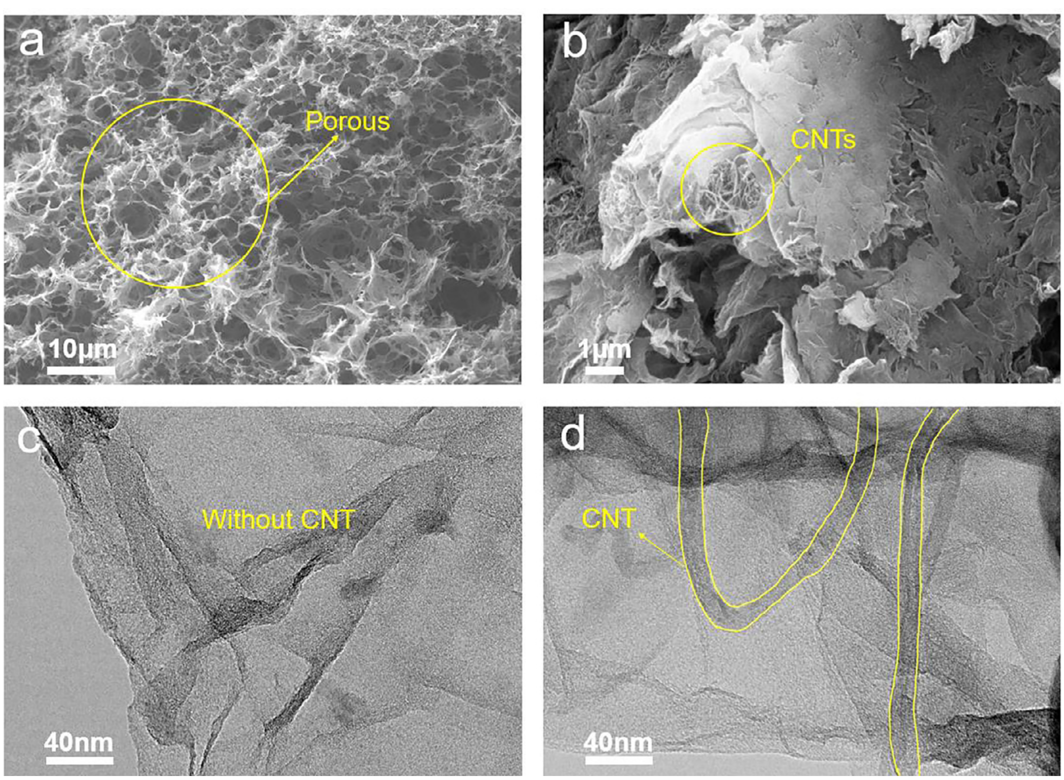

Fig. 3. (a-b) SEM images of CGA-10, (c) TEM image of GA, (d) TEM image of CGA-10.

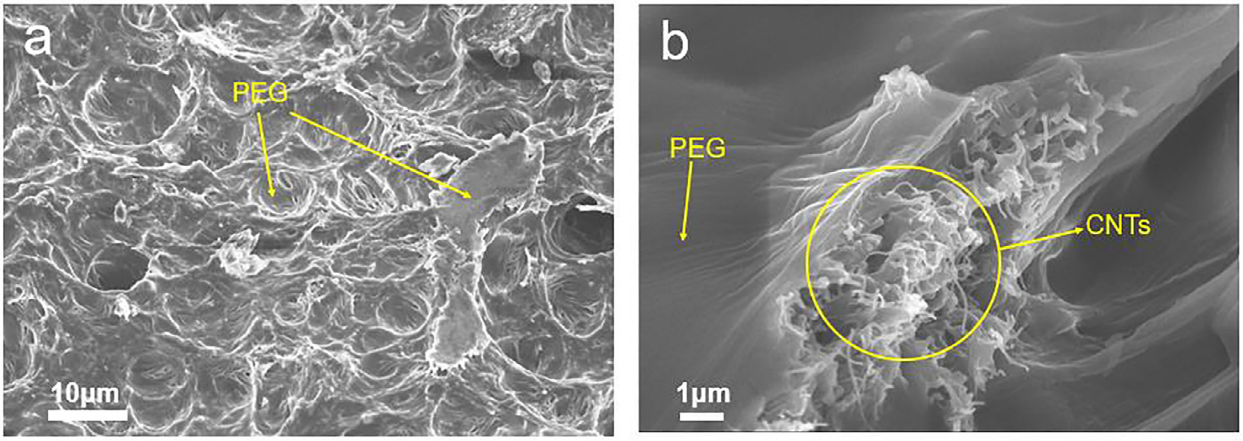

Fig. 4. (a-b) SEM images of PEG/CGA c-PCMs.

to investigate PEG/CGA c-PCMs. As we all know there are a large number of oxygen-containing functional groups on the surface of GO, such as carbonyl (COH), hydroxyl (OH), and carboxyl (COOH) [41,42]. A series of absorption peaks were visible from the FT-IR spectrum of the GO (Fig. 5(a)), demonstrated the existence of oxygen-containing functional groups. An obvious stretch vibration peak of –OH appeared at 3384 cm^{-1}.The stretch vibration peak of C═O and C═C emerged at 1729 cm^{-1} and 1619 cm^{-1} respectively which might be formed by the sp2 network structure of the GO crystal itself. Different characteristic peaks occurred at 1218 cm^{-1}, 1052 cm^{-1} and 846 cm^{-1}, representing the stretching vibration peak of C-OH, the stretching vibration peak of alkoxy C—O and the stretching vibration peak of C—O—C.

However, the above peaks existed in the infrared spectrum of GO hardly show up in the GA spectrum. It was caused by the disappearance of oxygen-containing functional groups on the GO surface during the hydrothermal reduction reaction and leaded to the formation of π-π bonds between the nanosheets, which was also the reason for the self-assembly of GO to form GA. Meanwhile it was also borne out in Raman spectra (Fig. 5(e-f)). The results exhibited that the R(R = I_D/I_G) of GO was 2.06, while the R(R = I_D/I_G) of CGA was 3.16. The increase of R value was attributed to the loss of oxygen-containing functional groups during the reduction process that caused the surface of the CGA to became rough and defects on the surface to increased.

The infrared spectra of PEG, CNT, GA, PEG/GA c-PCMs illustrates in Fig. 5(c). For PEG, the C—H stretching vibration peaks occurred at about 838 cm^{-1} and 946 cm^{-1}. Additionally, bending vibration peaks of C—H appeared among 1200 cm^{-1} to 1500 cm^{-1}.

The peaks located at 1118 cm^{-1} and 2884 cm^{-1} were attributed to the C—O stretch vibration and the stretch vibration of CH_2 respectively. Beyond that, OH peak appeared at 3434 cm^{-1}. For PEG/CGA c-PCMs and PEG/GA c-PCMs, they possessed the matching peaks, and these peaks contained all the peaks within PEG, CNT, and GA. All the peaks of PEG/CGA c-PCMs were the combination of PEG, CNT and GA, which indicated that there was no chemical reaction between PEG and CNT and GA. Similar phenomena also appeared in the XRD patterns (Fig. 5(d)). For PEG, there were two sharp characteristic peaks at 19° and 23°, which were also presented in PEG/CGA c-PCMs and PEG/GA c-PCMs. There were no additional peaks appeared in the pattern of PEG/CGA c-PCMs and PEG/GA c-PCMs. In the XRD patterns of GA and CNT (insert image of Fig. 5(d)), there was only one peak around 24.5°, which corresponded to the (002) crystal plane of graphite. The results of FT-IR and XRD confirmed that PEG, CNT and GA were only physical combination, and there was no chemical reaction between them.

3.3. Thermal properties of PEG/CGA c-PCMs

The phase transition behavior was evaluated by DSC, resulted in measurements of parameters such as phase transition temperature and enthalpy as illustrates in Fig. 6. In this study, PEG as a PCM was combined with the carrier CGA to formed a c-PCMs. PEG possessed the capacity to absorb and release energy, whereas the carrier CGA only participated in the composite process. Thus, the heat storage capacity of PEG/CGA c-PCMs was primarily reflected in the content of PEG. Fig. 6(a) presents the DSC curves of pure PEG, PEG/GA, and PEG/CGA. Only one absorption peak and release peak in accordance with the law of PEG solid–liquid phase transition were observed on the DSC curve, which also confirmed that PEG alone is involved in the energy absorption and release process during the phase transition.

According to the literature [43], the enthalpies of PEG or c-PCMs could be calculated by equation (2):

$$H = \frac{\Delta Q}{M} \quad (2)$$

where H is the enthalpy of PEG or c-PCMs, ΔQ presents the value of the heat change during the endothermic or exothermic process. M is the weight of PEG or c-PCMs. Measured results of phase transition temperature and enthalpy were summarized in Table 2.

The melting and solidifying temperature of pure PEG was 60.1 °C and 40.9 °C, respectively, and the actual melting and solidifying enthalpy was 174.8 J/g and 167.5 J/g. The melting and solidifying temperature of PEG/GA was 55.9 °C and 38.3 °C, and the actual melting and solidifying enthalpy was 155.9 J/g and 150.2 J/g, respectively. Meanwhile, PEG/CGA began to melt at 56.8 °C, with a melting enthalpy of 167.1 J/g. PEG/CGA started to solidified at 39.5 °C with a solidifying enthalpy of 162.9 J/g.

The melting and solidifying enthalpy of PEG/CGA c-PCMs and PEG/GA c-PCMs was slightly lower than that of pure PEG, which contributed to the fact that the presence of CNTs and GA reduced the relative mass of PEG in PEG/CGA c-PCMs, and the supporting material and fillers devoted no contribution to the latent heat capacities. In addition, compared to the pure PEG, the melting and solidifying temperature of PEG/CGA and PEG/GA decreased marginally. This might probably because PEG molecules entered into the three-dimensional porous structure of CGA, which restricted the phase transition thermodynamics of PEG [44]. Compared with PEG/GA c-PCMs, the melting temperature and solidifying temperature of PEG/CGA c-PCMs were slightly higher, which was contributed to the capillary action of CNTs and the reduced physical interaction between the PEG and CGA.

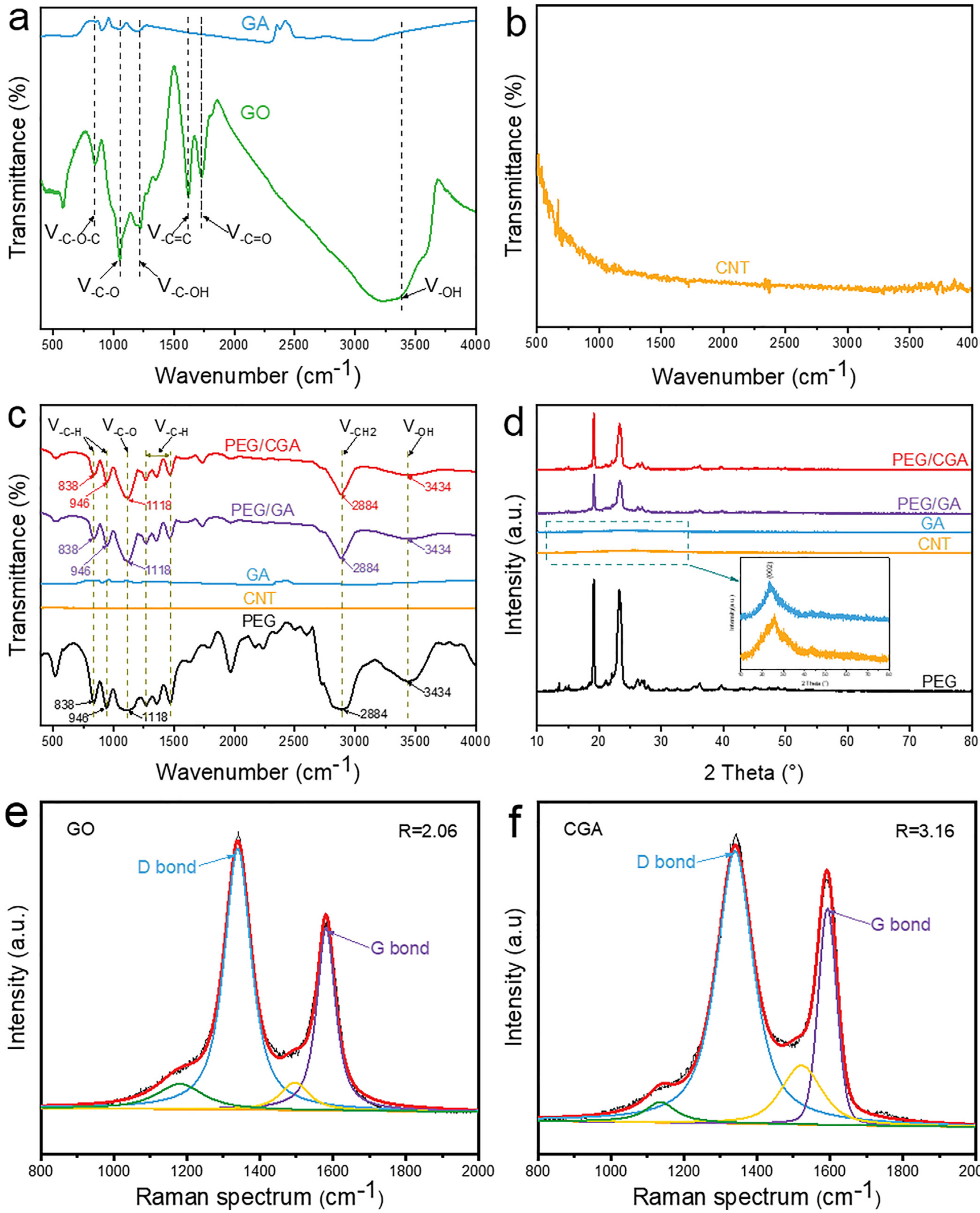

Fig. 5. FT-IR spectrum of (a) GA and GO, (b) CNT, (c) PEG, CNT, PEG/GA, and PEG/CGA c-PCMs. XRD pattern of (d) PEG, CNT, PEG/GA, and PEG/CGA c-PCMs. Raman spectrum of (e) GO, (f) CGA.

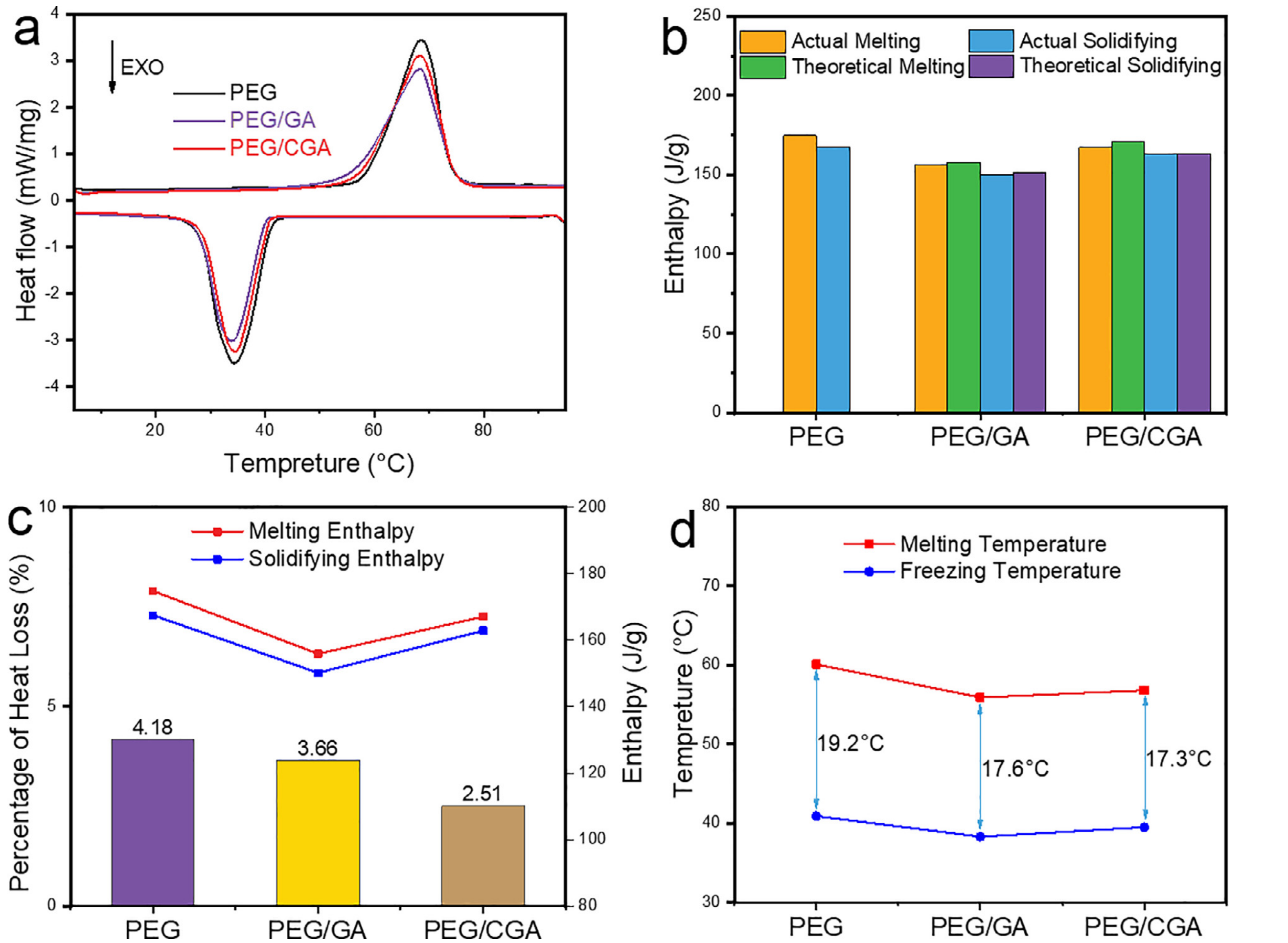

Fig. 6. Thermal characteristics of PEG,PEG/GA and PEG/CGA c-PCMs (a) DSC curves (b) actual enthalpy and theoretical enthalpy (c) percentage of heat loss and enthalpy (d) supercooling degree.

Table 2
Thermal characteristics of the PEG, PEG/GA, and PEG/CGA c-PCMs.

Samples	Melting process		Solidifying process	
	T_M (°C)	H_M (J/g)	T_S (°C)	H_S (J/g)
PEG	60.1	174.8	40.9	167.5
PEG/GA	55.9	155.9	38.3	150.2
PEG/CGA	56.8	167.1	39.5	162.9

Besides, according to the literature [39], the theoretical enthalpy of c-PCMs could be calculated by equation (3):

$$H_{theo} = \eta \cdot H_{PEG} \tag{3}$$

where H_{theo} is the theoretical enthalpy, η presents the mass fraction of the PEG in c-PCMs and H_{PEG} denotes the actual enthalpy of the PEG. All results were summarized in Table 3. As shown in Fig. 6(b), the actual enthalpy of c-PCMs was all slightly lower than the theoretical enthalpy. This might be caused by the resistance and steric hindrance effects of PEG after complexing with the porous three-dimensional structure carrier, which in turn inhibited the crystal arrangement and orientation of PEG single molecules that resulting in increased crystal disorder and defects [41].

The percentage of heat loss (W) used to evaluate the heat storage capacity is also important in c-PCMs, as calculated by equation (4) [45]:

$$W = (H_M - H_S)/H_M \times 100\% \tag{4}$$

where H_M and H_S are the melting and solidifying enthalpy of c-PCMs. In Fig. 6(c), the W values of PEG/CGA c-PCMs and PEG/GA c-PCMs were both lower than the pure PEG (4.18 %). Compared with PEG/GA c-PCMs (3.66 %), the heat loss percentage of PEG /CGA c-PCMs was reduced to 2.51 %, which because the highly porous structure of the CGA support and the capillary action of CNTs interfered with the crystallization of PEG [46]. Furthermore, the degree of undercooling (difference between melting and solidifying temperatures) is crucial in the application of c-PCMs [42]. It could be clearly seen from Fig. 6(d) that the undercooling degree of PEG/CGA c-PCMs was reduced from 19.2 °C in pristine PEG to 17.3 °C,

Table 3
Theoretical melting and solidifying enthalpy of PEG/GA and PEG/CGA c-PCMs.

Samples	PEG (wt%)	$H_{m,theo}$ (J/g)	$H_{s,theo}$ (J/g)
PEG/GA c-PCMs	90.2	157.7	151.0
PEG/CGA c-PCMs	97.6	170.6	163.4

which means that the addition of CNTs was beneficial to reduce the degree of undercooling.

We also compared the thermal storage capability of the as-prepared materials with previous work, including the melting and solidifying enthalpy. As illustrated in Table 4, compared with c-PCMs prepared by others, the PEG/CGA c-PCMs prepared in this work has the largest PEG mass fraction and the highest enthalpy of melting and solidifying. Therefore, this directly demonstrated that PEG/CGA c-PCMs has a large thermal storage capability.

3.4. Thermal stability of PEG/CGA c-PCMs

TGA was used to evaluate the thermal stability of PEG/CGA c-PCMs. All results were summarized in Table 5. As depicted in Fig. 7(a), the PEG started to lose weight at 187.9 °C and PEG reached a peak at 381.1 °C and finished at 437.7 °C. This indicated that the weight loss rate of PEG reached a maximum value at 381.1 °C. Likewise, PEG/CGA c-PCMs started to decompose at 212.5 °C and reached the maximum decomposition rate at 395.9 °C. Fig. 7(b) displays the weight loss rate of PEG was 99.63 %, and the weight loss rate of PEG/CGA c-PCMs was 97.49 % which mean the PEG accounted 97.49 wt% and CGA accounted 2.51 wt% in the PEG/CGA c-PCMs (PEG accounted 97.49 wt%, GA accounted 2.259 wt% and CNTs accounted 0.251 wt% in the PEG/CGA c-PCMs).

Since the temperature in practical application scenarios is always below 100 °C, which is much lower than the temperature corresponding to the maximum decomposition rate of PEG/CGA c-PCMs, so this indicates that the PEG/CGA c-PCMs could maintain its crystallinity during practical uses.

Table 4
Thermal properties comparison between PEG/CGA c-PCMs prepared in this work and some c-PCMs previously prepared in related references.

PCM	PCM (wt %)	H_M (J/g)	H_S (J/g)	Ref.
PEG	97.61	167.1	162.9	This work
PEG	66.38	134.93	129.27	[32]
PEG	51.70	59.25	65.45	[47]
paraffin	53.20	101.14	103.00	[44]
lauric acid	60.00	93.36	94.87	[48]
acid-stearic acid	72.20	117.30	114.50	[48]
capric-myristic acid	55.01	85.40	89.75	[49]
n-octadecane	80.65	142.00	126.5	[50]

3.5. Thermal cycling stability of PEG/CGA c-PCMs

In the repeated usage process, the stability of PEG/CGA c-PCMs should be considered. The corresponding c-PCMs were put on the lattice filter paper, and then placed on the oven with the temperature set at 80 °C for 20 min to ensure they were heated uniformly. Then, the c-PCMs were cooled down to room temperature under natural conditions. Moreover, the above heating and cooling cycles were repeated 500 and 1000 times, and the properties of PEG/CGA c-PCMs after the cycles were tested by XRD, FT-IR and DSC which were exhibited in Fig. 8(a-c).

After 500 and 1000 heating/cooling cycles, the characteristic peaks of the XRD patterns (Fig. 8(a)) of the PEG/CGA c-PCMs have not changed, which indicated that the PEG/CGA c-PCMs have not undergone chemical reactions during the heating/cooling cycles and also reflected the outstanding cycling stability of PEG/CGA c-PCMs. Similar results could also be obtained with FT-IR spectra (Fig. 8(b)), the characteristic absorption peaks appeared at the same positions in the PEG/CGA c-PCMs before and after the cycles. Meanwhile, as presented in Fig. 8(c), after 500 and 1000 heating/-cooling cycles, neither the endothermic nor the exothermic curves have changed significantly.

All results were summarized in Table 6. After 500 and 1000 repeated heating/cooling cycles, the T_M of the PEG/CGA c-PCMs changed by 0.3 °C and 0.2 °C, and the T_S by 0.4 °C and 0.3 °C, respectively. The results revealed that the variation range of the phase transition temperatures (T_M and T_S) of the PEG/CGA c-PCMs was negligible with the increase of the number of heating/-

Table 5
TGA data of the PEG and PEG/CGA c-PCMs.

Samples	Onset (°C)	Peak (°C)	Endset (°C)	Residual mass (%)
PEG	187.9	381.1	437.7	0.37
PEG/CGA c-PCMs	212.5	395.9	451.2	2.51

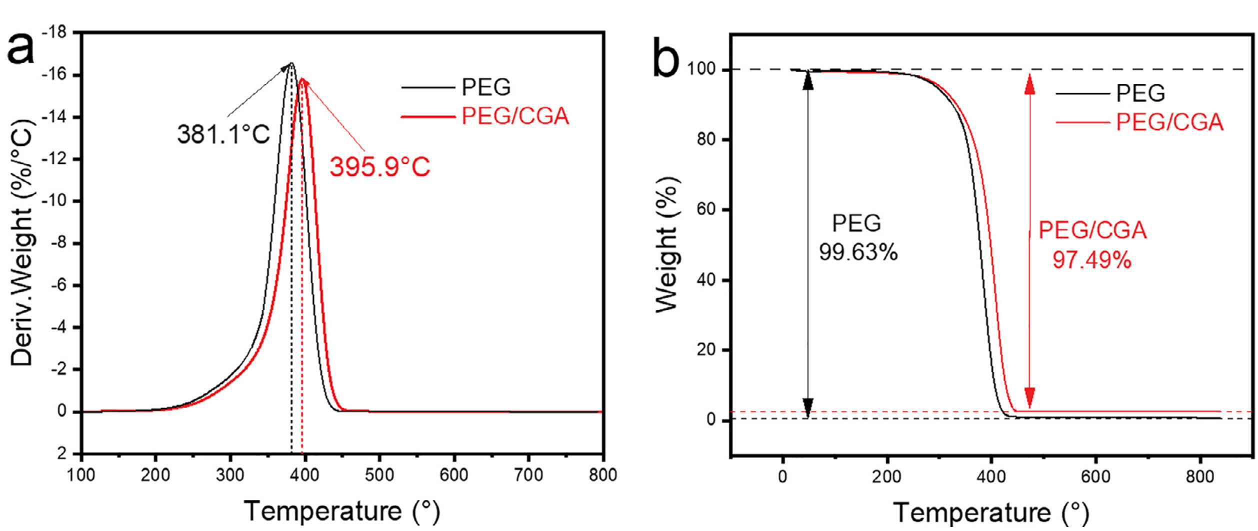

Fig. 7. (a) DTG and (b) TGA curves of PEG and PEG/CGA c-PCMs.

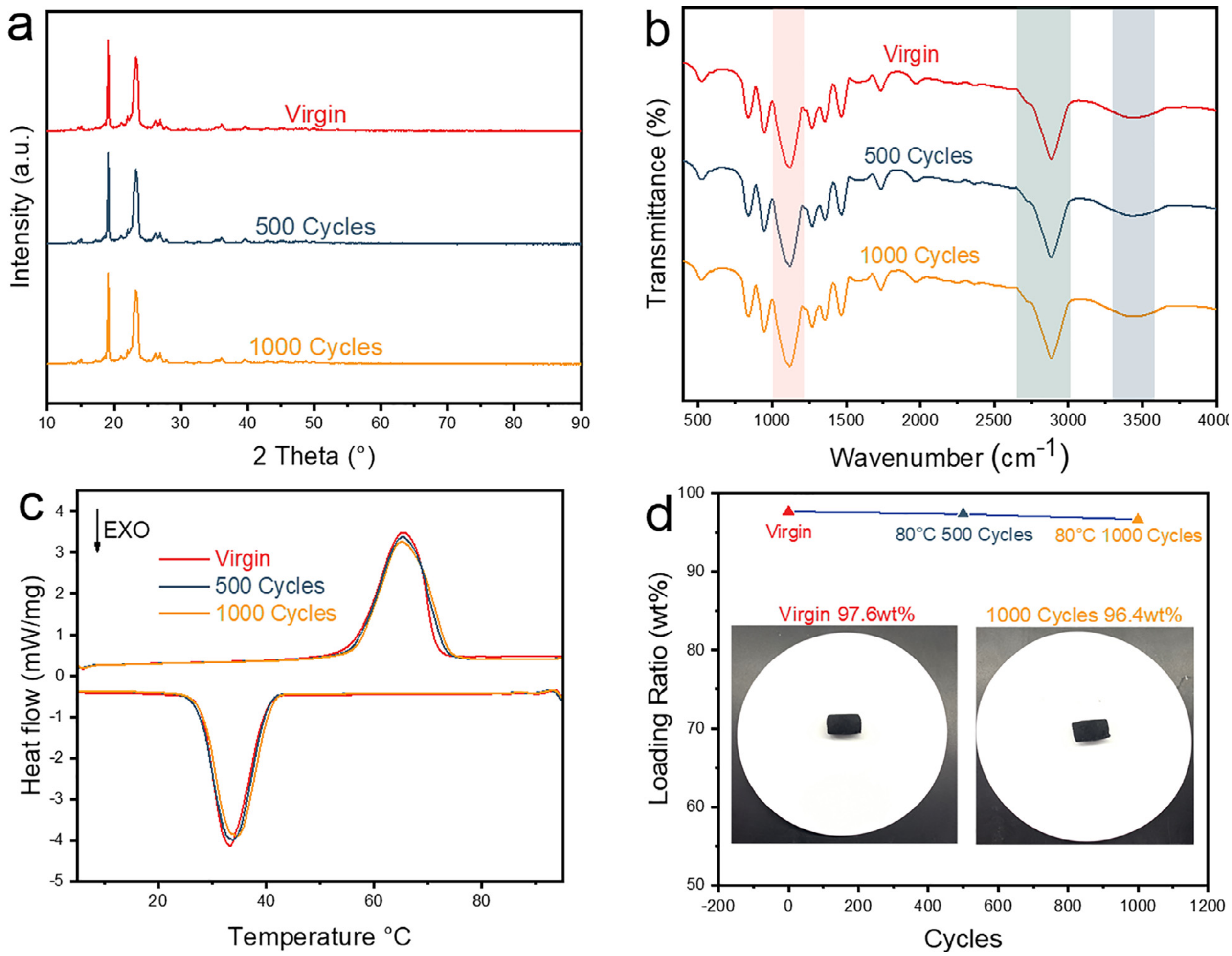

Fig. 8. (a) XRD patterns, (b) FT-IR spectra, (c) DSC curves, (d) Loading ratio of PEG/CGA c-PCMs after 500, 1000 heating/cooling cycles.

Table 6
Thermal characteristics of the PEG/CGA c-PCMs after 500 and 1000 cycles.

Samples	Melting process		Solidifying process	
	T_M (°C)	H_M (J/g)	T_S(°C)	H_S (J/g)
Virgin	56.8	167.1	39.5	162.9
500 Cycles	57.1	162.6	39.9	156.8
1000 Cycles	57.3	156.1	39.8	151.3

cooling cycles. Moreover, after 1000 repeated heating/cooling cycles, the H_M and H_S value of PEG/CGA c-PCMs remained 93.44 % and 92.84 % of virgin sample. The variation of the latent heat value of phase transition was acceptable in practical applications. Additionally, the mass fraction of PEG in the PEG/CGA c-PCMs after thermal cycling was tested. Fig. 8(d) presents the change curve of PEG mass fraction in PEG/CGA c-PCMs after 500 and 1000 cycles. The mass fraction of PEG still maintained a relatively high value under long-term cycling, which further demonstrated that PEG/CGA c-PCMs possessed well-behaved thermal cycling stability that was attributed to the physical interactions such as capillary force and surface tension between PEG and CGA.

Taken together, these proved that PEG/CGA c-PCMs presented reliable thermal cycling stability and could be used in thermal storage and thermal management systems for an extended period of time.

3.6. Heat transfer performance of PEG/CGA c-PCMs

The performance of heat transfer is a very important parameter for the thermal storage and release. Heat transfer performance during melting and solidifying of PEG, PEG/GA c-PCMs and PEG/CGA c-PCMs were investigated using a multi-record data logger and thermal imager. As illustrated in Fig. 9(a), PEG/CGA c-PCMs used less time (603 s) than PEG/GA c-PCMs(727 s) and pure PEG (813 s) when the temperature was increased from 30 °C to 70 °C. Likewise, when the temperature was lowered the same value, the time required for PEG/CGA c-PCMs was less than that required for PEG/GA c-PCMs and pure PEG. Furthermore, we also used a thermal imager to study the temperature change rate of PEG/CGA c-PCMs, PEG/GA c-PCMs and pure PEG when heated for the same time. As shown in Fig. 9(b), the temperature of the samples gradually increased with the increase of time. After the same period of

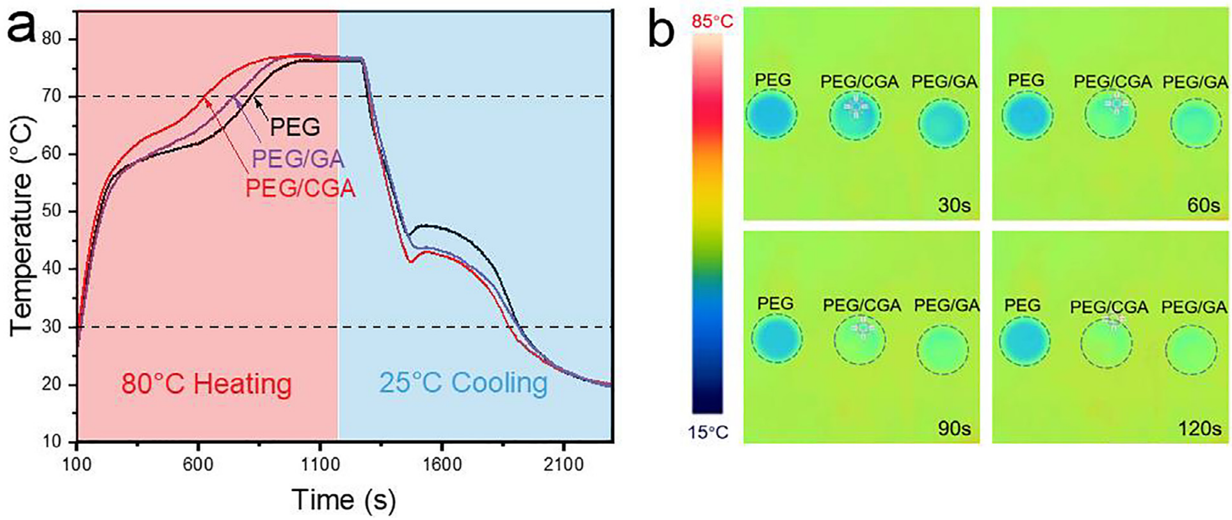

Fig. 9. (a-b) Heat transfer performance of PEG, PEG/GA and PEG/CGA c-PCMs.

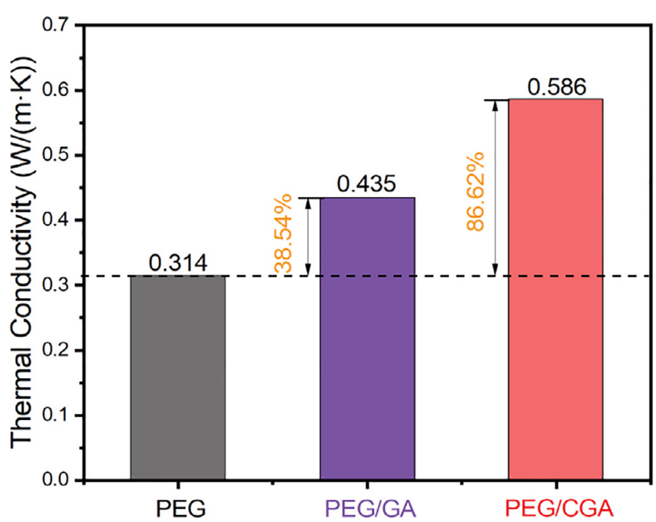

Fig. 10. Thermal conductivity of pure PEG, PEG/GA and PEG/CGA c-PCMs.

time, the temperature of PEG/CGA c-PCMs was the highest, followed by PEG/GA c-PCMs, while the temperature of pure PEG increased the slowest. This was an indication that the heat transfer efficiency of the PEG/CGA c-PCMs was the best which benefited from the high thermal conductivity of the GA framework and CNTs. It also proved that the addition of CNTs could improve the heat transfer performance of c-PCMs during thermal heat storage and release. These results specified that PEG/CGA c-PCMs could be used as remarkable and stable energy storage materials for environmental thermal management systems.

Thermal conductivity is a significant indicator affecting latent heat storage and release. Excellent thermal conductivity can respond to the change of external temperature rapidly to store or release heat timely. Fig. 10 displayed the thermal conductivity of PEG, PEG/GA c-PCMs and PEG/CGA c-PCMs. The thermal conductivities of the pure PEG, PEG/GA and PEG/CGA were 0.314, 0.435 and 0.586 W/(m·K), respectively. The thermal conductivity of PEG/GA was higher than that of pure PEG. GA, as a conduction channel, can enhance the thermal conductivity of pure PEG. Meanwhile, the thermal conductivity of PEG/CGA was 186 % times than that of pure PEG. CNTs with high aspect ratio, excellent thermal properties and large specific surface area are very important thermal conductive fillers. The thermal conductivity of the composites was further enhanced by the addition of CNTs owing to a secondary heat conduction network formed by the interweaving of CNTs and the heat transfer path was shortened.

4. Conclusion

In conclusion, we prepared PEG/CGA c-PCMs with excellent thermal stability and well-behaved heat transfer performance by GO self-assembly method and vacuum impregnation method. When the temperature was 200 °C and the doping amount of CNTs was 10 wt%, a porous three-dimensional CNTs-graphene hybrid aerogel with a large specific surface area was obtained. The incorporation of CNTs promoted the interaction between rGO nanosheets and improved the carrier's stability and adsorption to PEG. It was found by scanning electron microscopy that the PEG was uniformly distributed in the PEG/CGA c-PCMs, the mass fraction of PEG in the immersed PEG/CGA c-PCMs reached 97.6 wt% by calculation. The results of FT-IR and XRD indicated that no chemical reaction occurred between CGA and PEG. TG and DSC tests demonstrated that PEG/CGA c-PCMs have excellent thermal stability during practical use. After 500 and 1000 heating/cooling cycles, the H_M of PEG/CGA c-PCMs decreased by only 2.71 % and 3.79 % and the H_S by only 6.56 % and 7.16 % which proved that PEG/CGA c-PCMs presented reliable thermal cycling stability.

More importantly, the heat transfer performance of PEG/CGA c-PCMs is greatly improved due to the incorporation of CNTs and GA both with high thermal conductivity. The thermal conductivity of PEG/CGA was 186 % times than that of pure PEG. The test exposed that the heat transfer performance of PEG/CGA c-PCMs was the best, which greatly improved the heat transfer performance during latent heat storage and release. Therefore, the prepared PEG/CGA c-PCMs was considered to be the most promising material in the field of energy conservation and conversion.

Data availability

No data was used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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